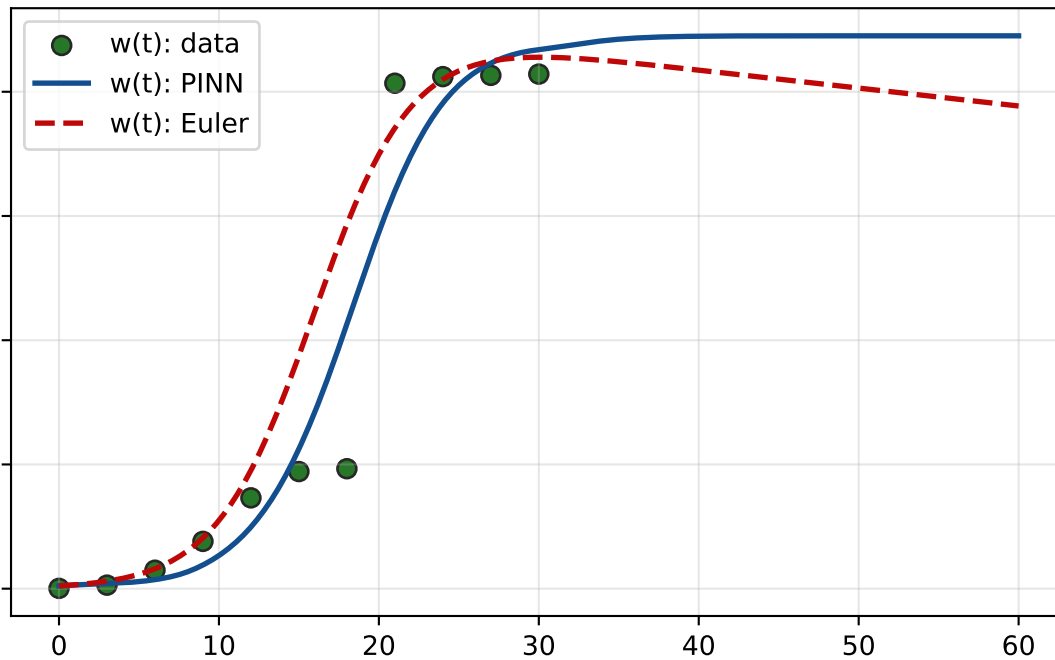
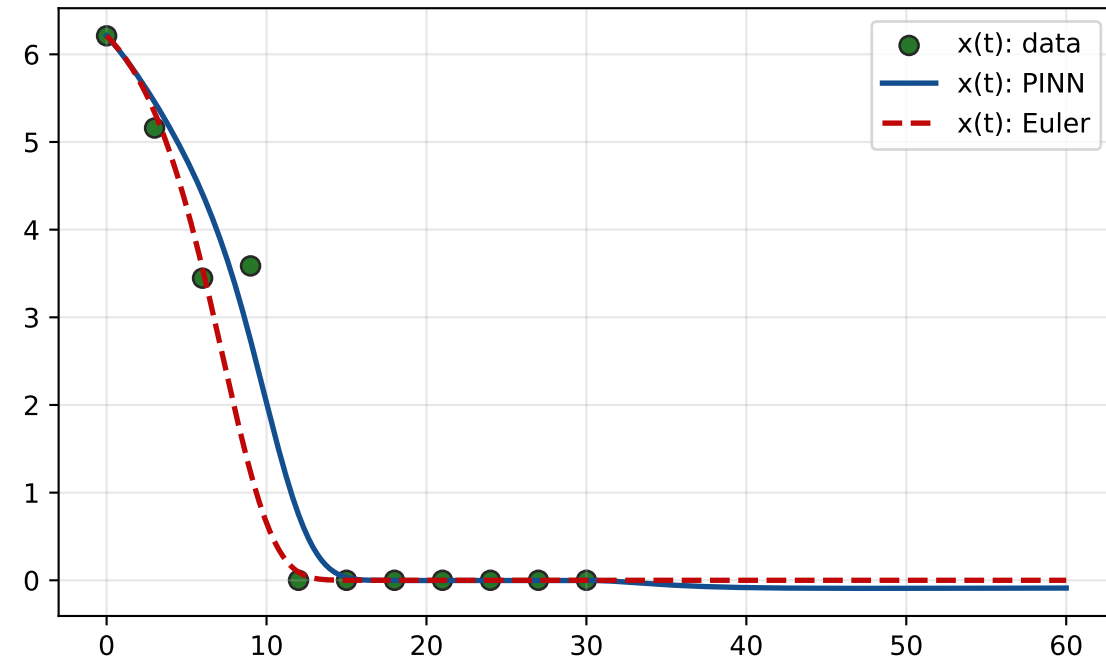


# E6 | PINN vs Euler

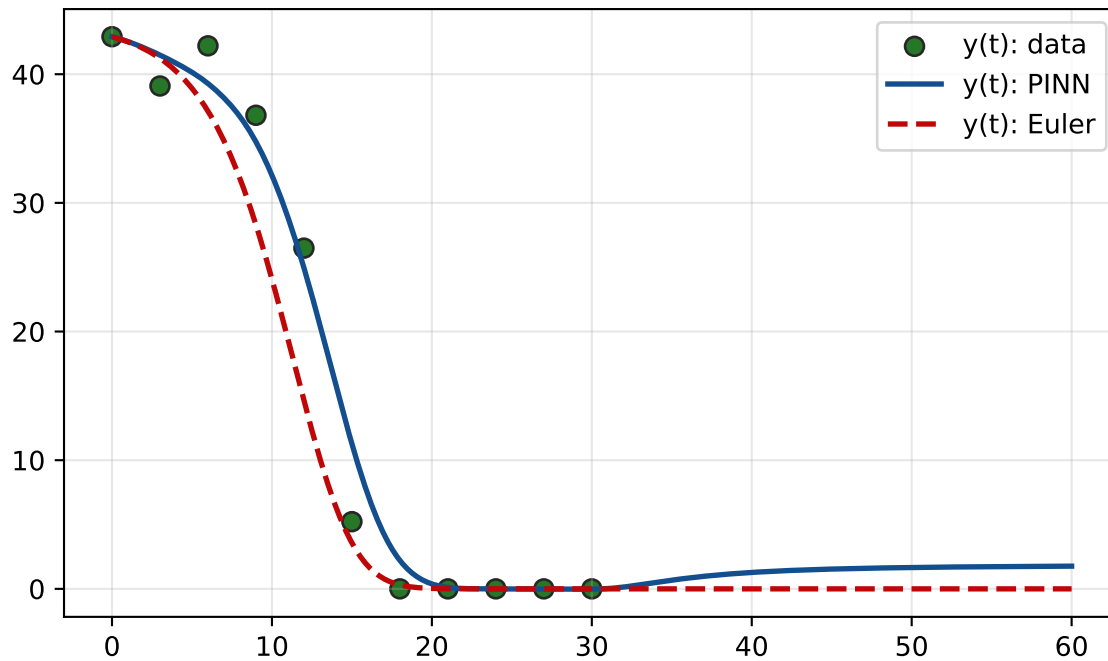
w(t): Biomass



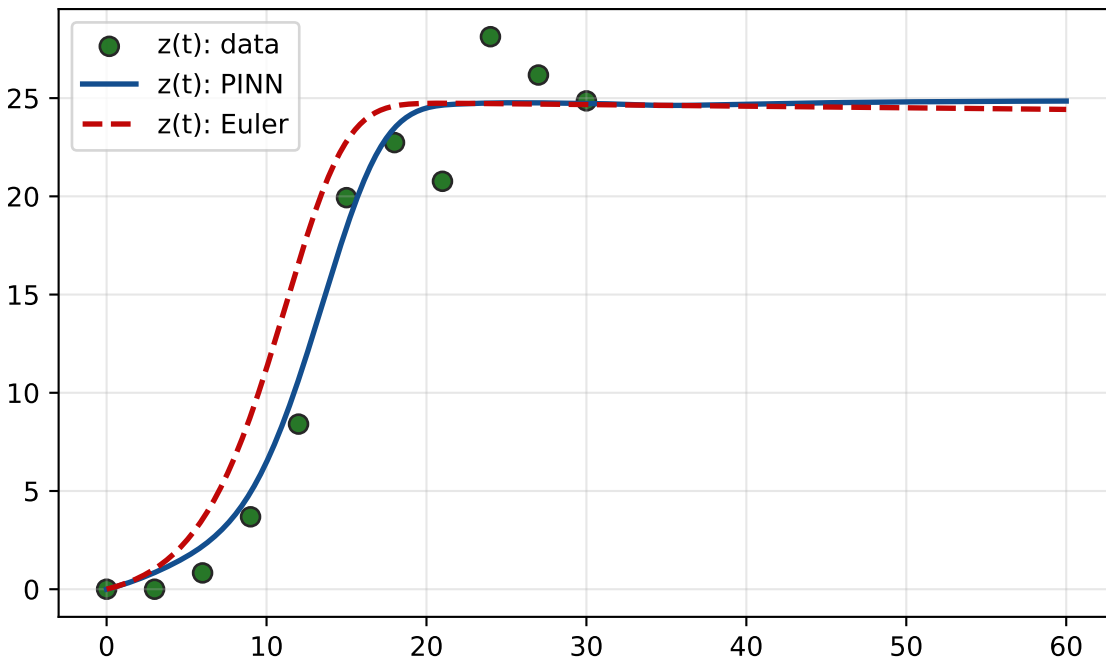
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E6

R2=0.9228, R2\_adj=0.9103, RSS=4.154e+02, AIC=1.131e+02

cond(JTJ)=4.38e+04, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 3.270447e-01 | 3.425735e-01 | 6.935037e-01 | -3.664591e-01 | 1.020548e+00 | 3.457785e-01 |
| p2    | 3.627679e-02 | 3.579670e-02 | 7.246662e-02 | -3.618983e-02 | 1.087434e-01 | 3.172730e-01 |
| p3    | 1.000000e-02 | 4.999463e-01 | 1.012088e+00 | -1.002088e+00 | 1.022088e+00 | 9.841463e-01 |
| p5    | 6.637883e-01 | 8.620955e-01 | 1.745221e+00 | -1.081433e+00 | 2.409009e+00 | 4.460777e-01 |
| p7    | 1.692028e-01 | 1.373148e-01 | 2.779794e-01 | -1.087766e-01 | 4.471822e-01 | 2.254352e-01 |
| p9    | 9.482867e-02 | 7.612142e-02 | 1.540998e-01 | -5.927109e-02 | 2.489284e-01 | 2.204806e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 4.434954e-02 |
| x0 (fixed)     | 6.211004e+00 |
| y0 (fixed)     | 4.291447e+01 |
| z0 (fixed)     | 0.000000e+00 |
| W0 (fixed)     | 0.000000e+00 |